Problems

Difficulty: ● Easy ■ Moderate ♦ Difficult ♦♦ Experts Only

13.1.1● For Example 13.4, define a set of random variables that could produce the sample function $m(t,s)$. Do not duplicate the set listed in Example 13.7.

13.1.2● For the random processes of Examples 13.3, 13.4, 13.5, and 13.6, identify whether the process is discrete-time or continuous-time, discrete-value or continuous-value.

13.1.3■ Let $Y(t)$ denote the random process corresponding to the transmission of one symbol over the QPSK communications system of Example 13.6. What is the sample space of the underlying experiment? Sketch the ensemble of sample functions.

13.1.4■ In a binary phase shift keying (BPSK) communications system, one of two equally probable bits, 0 or 1, is transmitted every T seconds. If the kth bit is $j \in \{0,1\}$, the waveform $x_j(t) = \cos(2\pi f_0 t + j\pi)$ is transmitted over the interval $[(k-1)T, kT]$. Let $X(t)$ denote the random process in which three symbols are transmitted in the interval $[0, 3T]$. Assuming f_0 is an integer multiple of $1/T$, sketch the sample space and corresponding sample functions of the process $X(t)$.

13.1.5■ True or false: For a continuous-value random process $X(t)$, the random variable $X(t_0)$ is always a continuous random variable.

13.2.1■ Let W be an exponential random variable with PDF

$$f_W(w) = \begin{cases} e^{-w} & w \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Find the CDF $F_{X(t)}(x)$ of the time-delayed ramp process $X(t) = t - W$.

13.2.2■ In a production line for 10 kHz oscillators, the output frequency of each oscillator is a random variable W uniformly distributed between 9980 Hz and 1020 Hz. The frequencies of different oscillators are independent. The oscillator company has an order for one part in 10^4 oscillators with frequency between 9999 Hz and 10,001 Hz. A technician takes one oscillator per minute from the production line and measures its exact frequency. (This test takes one minute.) The random variable T_r minutes is the elapsed time at which the technician finds r acceptable oscillators.

(a) Find p, the probability that an single oscillator has one-part-in-10^4 accuracy.

(b) What is $E[T_1]$ minutes, the expected time for the technician to find the first one-part-in-10^4 oscillator?

(c) What is the probability that the technician will find the first one-part-in-10^4 oscillator in exactly 20 minutes?

(d) What is $E[T_5]$, the expected time of finding the fifth one-part-in-10^4 oscillator?

13.2.3■ For the random process of Problem 13.2.2, what is the conditional PMF of T_2 given T_1? If the technician finds the first oscillator in 3 minutes, what is $E[T_2|T_1 = 3]$, the conditional expected value of the time of finding the second one-part-in-10^4 oscillator?

13.2.4♦ Let $X(t) = e^{-(t-T)}u(t-T)$ be an exponential pulse with a random delay T. The delay T has a PDF $f_T(t)$. Find the PDF of $X(t)$.

13.3.1● Suppose that at the equator, we can model the noontime temperature in degrees Celsius, X_n, on day n by a sequence of iid Gaussian random variables with expected value 30 degrees and standard deviation of 5 degrees. A new random process $Y_k = [X_{2k-1} + X_{2k}]/2$ is obtained by averaging the temperature over two days. Is Y_k an iid random sequence?

13.3.2■ For the equatorial noontime temperature sequence X_n of Problem 13.3.1, a second sequence of averaged temperatures is $W_n = [X_n + X_{n-1}]/2$. Is W_n an iid random sequence?

13.3.3■ Let Y_k denote the number of failures between successes $k-1$ and k of a Bernoulli (p) random process. Also, let Y_1 denote the number of failures before the first success. What is the PMF $P_{Y_k}(y)$? Is Y_k an iid random sequence?

13.4.1● The arrivals of new telephone calls at a telephone switching office is a Poisson process $N(t)$ with an arrival rate of $\lambda = 4$ calls per second. An experiment consists of monitoring the switching office and recording $N(t)$ over a 10-second interval.

(a) What is $P_{N(1)}(0)$, the probability of no phone calls in the first second of observation?

(b) What is $P_{N(1)}(4)$, the probability of exactly four calls arriving in the first second of observation?

(c) What is $P_{N(2)}(2)$, the probability of exactly two calls arriving in the first two seconds?

13.4.2● Queries presented to a computer database are a Poisson process of rate $\lambda = 6$ queries per minute. An experiment consists of monitoring the database for m minutes and recording $N(m)$, the number of queries presented. The answer to each of the following questions can be expressed in terms of the PMF $P_{N(m)}(k) = \mathrm{P}[N(m) = k]$.

(a) What is the probability of no queries in a one-minute interval?

(b) What is the probability of exactly six queries arriving in a one-minute interval?

(c) What is the probability of exactly three queries arriving in a one-half-minute interval?

13.4.3● At a successful garage, there is always a backlog of cars waiting to be serviced. The service times of cars are iid exponential random variables with a mean service time of 30 minutes. Find the PMF of $N(t)$, the number of cars serviced in the first t hours of the day.

13.4.4● The count of students dropping the course "Probability and Stochastic Processes" is known to be a Poisson process of rate 0.1 drops per day. Starting with day 0, the first day of the semester, let $D(t)$ denote the number of students that have dropped after t days. What is $P_{D(t)}(d)$?

13.4.5● Customers arrive at the Veryfast Bank as a Poisson process of rate λ customers per minute. Each arriving customer is immediately served by a teller. After being served, each customer immediately leaves the bank. The time a customer spends with a teller is called the service time. If the service time of a customer is exactly two minutes, what is the PMF of the number of customers $N(t)$ in service at the bank at time t?

13.4.6● Given a Poisson process $N(t)$, identify which of the following are Poisson processes..

(a) $N(2t)$, (b) $N(t/2)$,
(c) $2N(t)$, (d) $N(t)/2$,
(e) $N(t+2)$, (f) $N(t) - N(t-1)$.

13.4.7● Starting at any time t, the number N_τ of hamburgers sold at a White Castle in the τ minute interval $(t, t+\tau)$ has the Poisson PMF

$$P_{N_\tau}(n) = \begin{cases} (10\tau)^n e^{-10\tau}/n! & n = 0, 1, \ldots \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the expected number $\mathrm{E}[N_{60}]$ of hamburgers sold in one hour (60 minutes).

(b) What is the probability that no hamburgers are sold in the 10-minute interval starting at 12 noon?

(c) You arrive at the White Castle at 12 noon. You wait a random time W (minutes) until you see a hamburger sold. What is the PDF of W? Hint: Find $\mathrm{P}[W > w]$.

13.4.8■ A sequence of queries are made to a database system. The response time of the system, T seconds, is the exponential (1/8) random variable. As soon as the system responds to a query, the next query is made. Assuming the first query is made at time zero, let $N(t)$ denote the number of queries made by time t.

(a) What is $P[T \geq 4]$, the probability that a single query will last at least four seconds?

(b) If the database user has been waiting five seconds for a response, what is $P[T \geq 13|T \geq 5]$, the probability that the user will wait at least eight more seconds?

(c) What is the PMF of $N(t)$?

13.4.9■ The proof of Theorem 13.3 neglected the first interarrival time X_1. Show that X_1 has an exponential (λ) PDF.

13.4.10♦ $U_1, U_2, \ldots$ are independent identically distributed uniform random variables with parameters 0 and 1.

(a) Let $X_i = -\ln U_i$. What is $P[X_i > x]$?

(b) What kind of random variable is X_i?

(c) Given a constant $t > 0$, let N denote the value of n, such that

$$\prod_{i=1}^{n} U_i \geq e^{-t} > \prod_{i=1}^{n+1} U_i.$$

Note that we define $\prod_{i=1}^{0} U_i = 1$. What is the PMF of N?

13.5.1● Customers arrive at a casino as a Poisson process of rate 100 customers per hour. Upon arriving, each customer must flip a coin, and only those customers who flip heads actually enter the casino. Let $N(t)$ denote the process of customers entering the casino. Find the PMF of N, the number of customers who arrive between 5 PM and 7 PM.

13.5.2● A subway station carries both blue (B) line and red (R) line trains. Red line trains and blue line trains arrive as independent Poisson processes with rates $\lambda_R = 0.05$ and $\lambda_B = 0.15$ trains/min, respectively. You arrive at a random time t and wait until a red train arrives. Let N denote the number of blue line trains that pass through the station while you are waiting. What is the PMF $P_N(n)$?

13.5.3● A subway station carries both blue (B) line and red (R) line trains. Red line trains and blue line trains arrive as independent Poisson processes with rates $\lambda_R = 0.15$ and $\lambda_B = 0.30$ trains/min respectively. You arrive at the station at random time t and watch the trains for one hour.

(a) What is the PMF of N, the number of trains that you count passing through the station?

(b) Given that you see $N = 30$ trains, what is the conditional PMF of R, the number of red trains that you see?

13.5.4■ Buses arrive at a bus stop as a Poisson process of rate $\lambda = 1$ bus/minute. After a very long time t, you show up at the bus stop.

(a) Let X denote the interarrival time between two bus arrivals. What is the PDF $f_X(x)$?

(b) Let W equal the time you wait after time t until the next bus arrival. What is the PDF $f_W(w)$?

(c) Let V equal the time (in minutes) that has passed since the most recent bus arrival. What is the PDF $f_V(v)$?

(d) Let U equal the time gap between the most recent bus arrival and the next bus arrival. What is the PDF of U?

13.5.5■ For a Poisson process of rate λ, the Bernoulli arrival approximation assumes that in any very small interval of length Δ, there is either 0 arrivals with probability $1-\lambda\Delta$ or 1 arrival with probability $\lambda\Delta$. Use this approximation to prove Theorem 13.7.

13.5.6♦ Continuing Problem 13.4.5, suppose each service time is either one minute or two minutes equiprobably, independent of the arrival process or the other service times. What is the PMF of the number of customers $N(t)$ in service at the bank at time t?

13.5.7◆ Ten runners compete in a race starting at time $t = 0$. The runners' finishing times $R_1, \ldots, R_{10}$ are iid exponential random variables with expected value $1/\mu = 10$ minutes.

(a) What is the probability that the last runner will finish in less than 20 minutes?

(b) What is the PDF of X_1, the finishing time of the winning runner?

(c) Find the PDF of $Y = R_1 + \cdots + R_{10}$.

(d) Let $X_1, \ldots, X_{10}$ denote the runners' interarrival times at the finish line. Find the joint PDF $f_{X_1,\ldots,X_{10}}(x_1, \ldots, x_{10})$.

13.5.8◆◆ Let N denote the number of arrivals of a Poisson process of rate λ over the interval $(0, T)$. Given $N = n$, let $S_1, \ldots, S_n$ denote the corresponding arrival times. Prove that

$$f_{S_1,\ldots,S_n|N}(S_1, \ldots, S_n|n) = \begin{cases} n!/T^n & 0 \leq s_1 < \cdots < s_n \leq T, \\ 0 & \text{otherwise.} \end{cases}$$

Conclude that, given $N(T) = n$, $S_1, \ldots, S_n$ are the order statistics of a collection of n uniform $(0, T)$ random variables. (See Problem 5.10.11.)

13.6.1● Over the course of a day, the stock price of a widely traded company can be modeled as a Brownian motion process where $X(0)$ is the opening price at the morning bell. Suppose the unit of time t is an hour, the exchange is open for eight hours, and the standard deviation of the daily price change (the difference between the opening bell and closing bell prices) is 1/2 point. What is the Brownian motion parameter α?

13.6.2■ $X_0, X_1, \ldots$ is an iid Gaussian $(0, 1)$ random sequence. The random sequence Y_n is defined by $Y_0 = 0$ and $Y_{n+1} = X_{n+1} + Y_n$. Find the autocorrelation function $R_Y[n, k]$.

13.6.3■ Let $X(t)$ be a Brownian motion process with variance $\text{Var}[X(t)] = \alpha t$. For a constant $c > 0$, determine whether $Y(t) = X(ct)$ is a Brownian motion process.

13.6.4◆ For a Brownian motion process $X(t)$, let $X_0 = X(0), X_1 = X(1), \ldots$ represent samples of a Brownian motion process with variance αt. The discrete-time continuous-value process $Y_1, Y_2, \ldots$ defined by $Y_n = X_n - X_{n-1}$ is called an *increments process*. Show that Y_n is an iid random sequence.

13.6.5◆ This problem works out the missing steps in the proof of Theorem 13.8. For $\mathbf{W}$ and $\mathbf{X}$ as defined in the proof of the theorem, show that $\mathbf{W} = \mathbf{AX}$. What is the matrix $\mathbf{A}$? Use Theorem 8.11 to find $f_{\mathbf{W}}(\mathbf{w})$.

13.7.1● X_n is an iid random sequence with expected value $\text{E}[X_n] = \mu_X$ and variance $\text{Var}[X_n] = \sigma_X^2$. What is the autocovariance $C_X[m, k]$?

13.7.2■ For the time-delayed ramp process $X(t)$ from Problem 13.2.1, find for any $t \geq 0$:

(a) The expected value function $\mu_X(t)$,

(b) The autocovariance function $C_X(t, \tau)$. Hint: $\text{E}[W] = 1$ and $\text{E}[W^2] = 2$.

13.7.3■ A simple model (in degrees Celsius) for the daily temperature process $C(t)$ of Example 13.3 is

$$C_n = 16\left[1 - \cos\frac{2\pi n}{365}\right] + 4X_n,$$

where $X_1, X_2, \ldots$ is an iid random sequence of Gaussian $(0, 1)$ random variables.

(a) What is $\text{E}[C_n]$?

(b) Find the autocovariance function $C_C[m, k]$.

(c) Why is this model overly simple?

13.7.4◆ A different model for the daily temperature process $C(n)$ of Example 13.3 is

$$C_n = \frac{1}{2}C_{n-1} + 4X_n,$$

where $C_0, X_1, X_2, \ldots$ is an iid random sequence of Gaussian $(0,1)$ random variables.

(a) Find the mean and variance of C_n.

(b) Find the autocovariance $C_C[m,k]$.

(c) Is this a plausible model for the daily temperature over the course of a year?

(d) Would $C_1, \ldots, C_{31}$ constitute a plausible model for the daily temperature for the month of January?

13.7.5■ For a Poisson process $N(t)$ of rate λ, show that for $s < t$, the autocovariance is $C_N(s,t) = \lambda s$. If $s > t$, what is $C_N(s,t)$? Is there a general expression for $C_N(s,t)$?

13.7.6♦ $N(t)$ is a Poisson process of rate $\lambda = 1$ and $X_0, X_1, X_2, \ldots$ is an iid sequence of Gaussian $(0, \sigma)$ random variables that are independent of $N(t)$. Consider the process $\{Y(t)|t \geq 0\}$ defined by $Y(t) = X_{N(t)}$. Find the expected value $\mu_Y(t) = \mathrm{E}[Y(t)]$ and the covariance function $C_Y(t,\tau)$. (Assume $|\tau| < t$.)

13.7.7♦ X_n is an iid random sequence with $\mathrm{E}[X_n] = 0$ and $\mathrm{Var}[X_n] = 3$. Find the autocorrelation function $C_Y[n,k]$ of the process $Y_n = X_{n-1}X_n$.

13.8.1● For an arbitrary constant a, let $Y(t) = X(t+a)$. If $X(t)$ is a stationary random process, is $Y(t)$ stationary?

13.8.2● $\mathbf{X} = \begin{bmatrix} X_1 & X_2 \end{bmatrix}'$ has expected value $\mathrm{E}[\mathbf{X}] = \mathbf{0}$ and covariance matrix

$$\mathbf{C}_X = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

Does there exist a stationary process $X(t)$ and time instances t_1 and t_2 such that $\mathbf{X}$ is actually a pair of observations $\begin{bmatrix} X(t_1) & X(t_2) \end{bmatrix}'$ of the process $X(t)$?

13.8.3● For an arbitrary constant a, let $Y(t) = X(at)$. If $X(t)$ is a stationary random process, is $Y(t)$ stationary?

13.8.4● Let $X(t)$ be a stationary continuous-time random process. By sampling $X(t)$ every Δ seconds, we obtain the discrete-time random sequence $Y_n = X(n\Delta)$. Is Y_n a stationary sequence?

13.8.5● Given a stationary random sequence X_n, we can *subsample* X_n by extracting every kth sample: $Y_n = X_{kn}$. Is Y_n a stationary random sequence?

13.8.6■ Let A be a nonnegative random variable that is independent of any collection of samples $X(t_1), \ldots, X(t_k)$ of a stationary random process $X(t)$. Is $Y(t) = AX(t)$ a stationary random process?

13.8.7♦ Let $g(x)$ be deterministic function. If $X(t)$ is a stationary random process, is $Y(t) = g(X(t))$ a stationary process?

13.9.1● Which of the following are valid autocorrelation functions?

$$R_1(\tau) = \delta(\tau) \qquad R_2(\tau) = \delta(\tau) + 10$$
$$R_3(\tau) = \delta(\tau - 10) \qquad R_4(\tau) = \delta(\tau) - 10$$

13.9.2● Let A be a nonnegative random variable that is independent of any collection of samples $X(t_1), \ldots, X(t_k)$ of a wide sense stationary random process $X(t)$. Is $Y(t) = A + X(t)$ a wide sense stationary process?

13.9.3● True or False: If X_n is a wide sense stationary random sequence with $\mathrm{E}[X_n] = 0$, then $Y_n = X_n - X_{n-1}$ is a wide sense stationary random sequence.

13.9.4● Let X_n denote an iid sequence of Bernoulli $(p = 1/2)$ random variables. Find the autocorrelation function $R_X[n,k]$ and the autocovariance function $C_X[n,k]$.

13.9.5● X_n is an iid sequence with $\mathrm{E}[X_n] = \mu$ and $\mathrm{Var}[X_n] = \sigma^2$. Find the autocorrelation function $R_X[n,k]$.

13.9.6■ $X(t)$ and $Y(t)$ are independent wide sense stationary processes. Determine if these processes are wide-sense stationary:

(a) $V(t) = X(t) + Y(t)$,

(b) $W(t) = X(t)Y(t)$.

13.9.7■ True or False: If X_n is a wide sense stationary random sequence with $E[X_n] = 0$, then $Y_n = X_n + (-1)^{n-1}X_{n-1}$ is a wide sense stationary random sequence.

13.9.8■ Consider the random process

$$W(t) = X\cos(2\pi f_0 t) + Y\sin(2\pi f_0 t),$$

where X and Y are uncorrrelated random variables, each with expected value 0 and variance σ^2. Find the autocorrelation $R_W(t,\tau)$. Is $W(t)$ wide sense stationary?

13.9.9■ $X(t)$ is a wide sense stationary random process with average power equal to 1. Let Θ denote a random variable with uniform distribution over $[0, 2\pi]$ such that $X(t)$ and Θ are independent.

(a) What is $E[X^2(t)]$?

(b) What is $E[\cos(2\pi f_c t + \Theta)]$?

(c) Let $Y(t) = X(t)\cos(2\pi f_c t + \Theta)$. What is $E[Y(t)]$?

(d) What is the average power of $Y(t)$?

13.9.10■ Prove the properties of $R_X[n]$ given in Theorem 13.12.

13.9.11◆ Let X_n be a wide sense stationary random sequence with expected value μ_X and autocovariance $C_X[k]$. For $m = 0, 1, \ldots,$ we define

$$\overline{X}_m = \frac{1}{2m+1}\sum_{n=-m}^{m} X_n$$

as the sample mean process. Prove that if $\sum_{k=-\infty}^{\infty} C_X[k] < \infty$, then $\overline{X}_0, \overline{X}_1, \ldots$ is an unbiased consistent sequence of estimates of μ_X.

13.9.12◆ Determine whether each of these statements is true or false:

(a) If X_n and Y_n are independent stationary processes, then $V_n = X_n/Y_n$ is wide-sense stationary.

(b) If X_n and Y_n are independent wide sense stationary processes, then $W_n = X_n/Y_n$ is wide-sense stationary.

13.10.1● $X(t)$ and $Y(t)$ are independent wide sense stationary processes with expected values μ_X and μ_Y and autocorrelation functions $R_X(\tau)$ and $R_Y(\tau)$, respectively. Let $W(t) = X(t)Y(t)$.

(a) Find μ_W and $R_W(t,\tau)$ and show that $W(t)$ is wide sense stationary.

(b) Are $W(t)$ and $X(t)$ jointly wide sense stationary?

13.10.2■ $X(t)$ is a wide sense stationary random process. For each process $X_i(t)$ defined below, determine whether $X_i(t)$ and $X(t)$ are jointly wide sense stationary.

(a) $X_1(t) = X(t+a)$

(b) $X_2(t) = X(at)$

13.10.3■ $X(t)$ is a wide sense stationary stochastic process with autocorrelation function

$$R_X(\tau) = 10\sin(2\pi 1000\tau)/(2\pi 1000\tau).$$

The process $Y(t)$ is a version of $X(t)$ delayed by 50 microseconds: $Y(t) = X(t - t_0)$ where $t_0 = 5\times 10^{-5}$s.

(a) Derive the autocorrelation function of $Y(t)$.

(b) Derive the cross-correlation function of $X(t)$ and $Y(t)$.

(c) Is $Y(t)$ wide-sense stationary?

(d) Are $X(t)$ and $Y(t)$ jointly wide sense stationary?

13.11.1● A stationary Gaussian process $X(t)$ is observed at times t_1 and t_2 to form the random vector $\mathbf{X} = \begin{bmatrix} X(t_1) & X(t_2)\end{bmatrix}'$ with expected value $E[\mathbf{X}] = \mathbf{0}$ and covariance matrix $\mathbf{C}_X = \begin{bmatrix}\sigma_1^2 & 1\\ 1 & \sigma_2^2\end{bmatrix}$. What is the range of valid values (if any) of σ_1^2 and σ_2^2?

13.11.2■ Given a Gaussian process $X(t)$, identify which of the following, if any, are Gaussian processes.

(a) $2X(t)$,

(b) $X(t/2)$,

(c) $X(t)/2$,

(d) $X(t) - X(t-1)$,

(e) $X(2t)$.

13.11.3■ A white Gaussian noise process $N(t)$ with autocorrelation $R_N(\tau) = \alpha\delta(\tau)$ is passed through an integrator yielding the output

$$Y(t) = \int_0^t N(u)\,du.$$

Find $E[Y(t)]$ and the autocorrelation function $R_Y(t,\tau)$. Show that $Y(t)$ is a nonstationary process.

13.11.4■ Let $X(t)$ be a Gaussian process with mean $\mu_X(t)$ and autocovariance $C_X(t,\tau)$. In this problem, we verify that the for two samples $X(t_1), X(t_2)$, the multivariate Gaussian density reduces to the bivariate Gaussian PDF. In the following steps, let σ_i^2 denote the variance of $X(t_i)$ and let $\rho = C_X(t_1, t_2 - t_1)/(\sigma_1\sigma_2)$ equal the correlation coefficient of $X(t_1)$ and $X(t_2)$.

(a) Find the covariance matrix $\mathbf{C}$ and show that the determinant is $|\mathbf{C}| = \sigma_1^2\sigma_2^2(1-\rho^2)$.

(b) Show that the inverse of the correlation matrix is

$$\mathbf{C}^{-1} = \frac{1}{1-\rho^2}\begin{bmatrix} \frac{1}{\sigma_1^2} & \frac{-\rho}{\sigma_1\sigma_2} \\ \frac{-\rho}{\sigma_1\sigma_2} & \frac{1}{\sigma_1^2} \end{bmatrix}.$$

(c) Now show that the multivariate Gaussian density for $X(t_1), X(t_2)$ is the bivariate Gaussian density.

13.11.5■ Show that the Brownian motion process is a Gaussian random process. Hint: For $\mathbf{W}$ and $\mathbf{X}$ as defined in the proof of the Theorem 13.8, find matrix $\mathbf{A}$ such that $\mathbf{W} = \mathbf{AX}$ and then apply Theorem 8.11.

13.11.6♦♦ Let $X_1, X_2, \ldots$ denote a sequence of iid Gaussian $(0,1)$ random variables. Let $N(t)$ denote a Poisson process of rate λ that is independent of the sequence X_n. Consider the random process

$$Y(t) = \sum_{n=0}^{N(t)} X_n.$$

(a) Find the conditional CDF

$$F_{Y(t)|N(t)}(y|n) = P[Y(t) \le y | N(t) = n].$$

Express your answer in terms of the $\Phi(\cdot)$ function.

(b) Is $Y(t)$ a Gaussian process?

(c) Is $Y(t)$ a stationary process?

(d) Is $Y(t)$ wide-sense stationary?

13.12.1● Write a MATLAB program that generates and graphs the noisy cosine sample paths $X_{cc}(t)$, $X_{dc}(t)$, $X_{cd}(t)$, and $X_{dd}(t)$ of Figure 13.3. Note that the mathematical definition of $X_{cc}(t)$ is

$$X_{cc}(t) = 2\cos(2\pi t) + N(t), \quad -1 \le t \le 1.$$

Note that $N(t)$ is a white noise process with autocorrelation $R_N(\tau) = 0.01\delta(\tau)$. Practically, the graph of $X_{cc}(t)$ in Figure 13.3 is a sampled version $X_{cc}[n] = X_{cc}(nT_s)$, where the sampling period is $T_s = 0.001$s. In addition, the discrete-time functions are obtained by subsampling $X_{cc}[n]$. In subsampling, we generate $X_{dc}[n]$ by extracting every kth sample of $X_{cc}[n]$; see Problem 13.8.5. In terms of MATLAB, which starts indexing a vector `x` with first element `x(1)`,

```
Xdc(n)=Xcc(1+(n-1)k).
```

The discrete-time graphs of Figure 13.3 used $k = 100$.

13.12.2● For the telephone switch of Example 13.28, we can estimate the expected number of calls in the system, $E[M(t)]$, after T minutes using the time average estimate

$$\overline{M}_T = \frac{1}{T}\sum_{k=1}^{T} M(k).$$

Perform a 600-minute switch simulation and graph the sequence $\overline{M}_1, \overline{M}_2, \ldots, \overline{M}_{600}$. Does it appear that your estimates are converging? Repeat your experiment ten times and interpret your results.

13.12.3● A particular telephone switch handles only automated junk voicemail calls that arrive as a Poisson process of rate

$\lambda = 100$ calls per minute. Each automated voicemail call has duration of exactly one minute. Use the method of Problem 13.12.2 to estimate the expected number of calls $\mathrm{E}[M(t)]$. Do your results differ very much from those of Problem 13.12.2?

13.12.4■ Recall that for a rate λ Poisson process, the expected number of arrivals in $[0, T]$ is λT. Inspection of the code for `poissonarrivals(lambda,T)` will show that initially $n = \lceil 1.1\lambda T \rceil$ arrivals are generated. If $S_n > T$, the program stops and returns $\{S_j | S_j \leq T\}$. Otherwise, if $S_n < T$, then we generate an additional n arrivals and check if $S_{2n} > T$. This process may be repeated an arbitrary number of times k until $S_{kn} > T$. Let K equal the number of times this process is repeated. What is $\mathrm{P}[K = 1]$? What is the disadvantage of choosing larger n so as to increase $\mathrm{P}[K = 1]$?

13.12.5■ In this problem, we employ the result of Problem 13.5.8 as the basis for a function `s=newarrivals(lambda,T)` that generates a Poisson arrival process. The program `newarrivals.m` should do the following:

- Generate a sample value of N, a Poisson (λT) random variable.
- Given $N = n$, generate $\{U_1, \ldots, U_n\}$, a set of n uniform $(0, T)$ random variables.
- Sort $\{U_1, \ldots, U_n\}$ from smallest to largest and return the vector of sorted elements.

Write the program `newarrivals.m` and experiment to find out whether this program is any faster than `poissonarrivals.m`.

13.12.6♦ Suppose the Brownian motion process with $\alpha = 1$ is constrained by barriers. That is, we wish to generate a process $Y(t)$ such that $-b \leq Y(t) \leq b$ for a constant $b > 0$. Build a simulation of this system. Estimate $\mathrm{P}[Y(t) = b]$.

13.12.7♦ For the departure process $D(t)$ of Example 13.28, let D_n denote the time of the nth departure. The nth *interdeparture time* is then $V_n = D_n - D_{n-1}$. From a sample path containing 1000 departures, estimate the PDF of V_n. Is it reasonable to model V_n as an exponential random variable? What is the mean interdeparture time?